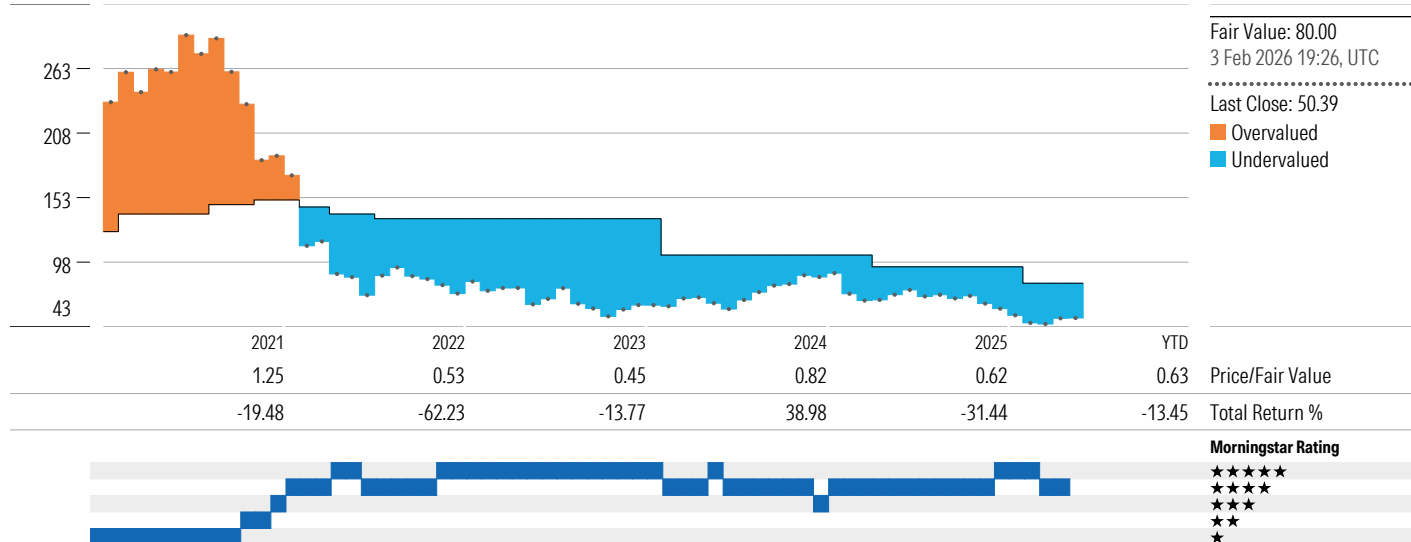


PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
50.39 USD 4 May 2026	80.00 USD 4 Feb 2026 01:26, UTC	0.63	41.33 USD Bil 5 May 2026	Narrow	Mid Value	High	Standard	5 May 2026 05:00, UTC

Price vs. Fair Value



Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 3 Feb 2026 19:26, UTC.

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Appendix

Research Methodology for Valuing Companies

Important Disclosure

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The primary analyst covering this company does not own its stock.

¹The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

PayPal Earnings: Muted Results as New CEO Takes the Reins

Analyst Note Brett Horn, CFA, Senior Equity Analyst, 5 May 2026

PayPal's first-quarter results were weak in an absolute sense but were ahead of management's expectations as the company resets. However, we think the market may be reacting negatively to some warning signs of macro pressure and the new CEO's strategy.

Why it matters: Net revenue increased 5% year over year on a constant-currency basis, and transaction margin dollars (essentially gross profit) were up 3%.

- While top-line results were modestly encouraging, adjusted margins declined to 18.4% from 20.7% last year as the company makes investments for growth.
- Overall payment volume growth was solid at 8%. PSP (Braintree) saw volume growth accelerate modestly to 11% during the first quarter from 8% last quarter. PayPal branded volume came in at 5%, a slight sequential improvement. However, management pointed to some issues with travel spending so far in the second quarter.

The bottom line: We will maintain our \$80 per share fair value estimate for the narrow-moat company and see the shares as materially overvalued.

- PayPal is exposed to consumer spending, and online volumes tend to skew toward discretionary purchases. If the macro situation worsens materially, management's efforts to stabilize the business and return to growth could become more difficult. However, we think a market price that equates to about a 15% yield on expected 2026 adjusted free cash flow more than compensates for this risk.

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Sector	Industry
Financial Services	Credit Services

Business Description

PayPal was spun off from eBay in 2015 and provides electronic payment solutions to merchants and consumers, with a focus on online transactions. The company had 439 million active accounts at the end of 2025. The company also owns Venmo, a person-to-person payment platform.

- Outside of results, new CEO Enrique Loes provided some color on his plans, which include updating the company's technology platforms to restore competitiveness. We think this move makes sense but will likely take some time, and the market may be disappointed by the lack of a quick fix.

Business Strategy & Outlook Brett Horn, CFA, Senior Equity Analyst, 10 Apr 2026

PayPal's early development of a network of merchants and consumers enabled the company to build and maintain a strong competitive position in the online payments channel. Historically, PayPal's growth had been driven by the ongoing shift toward electronic payments and the rise of e-commerce, which the coronavirus pandemic temporarily accelerated. However, the company has faced headwinds as the pandemic-related positives reversed and new competition emerged. Management has sought to counter pressure on top-line growth by increasing focus on cost control and product innovation, with the goal of shifting to more profitable growth.

We see this evolution as the right move, but it will likely take some time to fully execute. Recently, the company announced a new CEO, as the board felt that progress had been too slow. As a result, 2026 will likely be a transition year.

In the long term, we see a mix of competitive opportunities and threats that could lead to a wide range of outcomes. PayPal remains a somewhat unique player within the payments space. We think this remains its key strength, but its position on both the merchant and consumer side could be increasingly challenged over the long run. On the merchant side, new online-focused acquirers have emerged, and financial technology innovation also appears to be concentrated in the e-commerce space. On the consumer side, services such as Apple Pay represent competition for PayPal. Competition on both sides could chip away at PayPal's position. On the other hand, PayPal remains a preferred partner in the online space, its Braintree business is becoming more profitable and we think Venmo creates option value. In balance, we think the company can hold its own, but we recognize the potential to veer in either direction.

PayPal's online focus leaves it relatively exposed to an economic downturn, as online purchases tend to be discretionary. If the macroeconomic picture worsens, it could put more pressure on the company and obscure any underlying improvements.

Bulls Say Brett Horn, CFA, Senior Equity Analyst, 10 Apr 2026

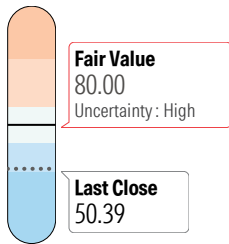
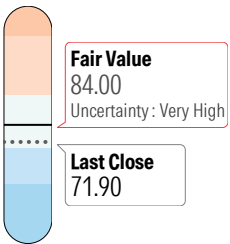
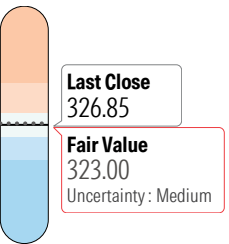
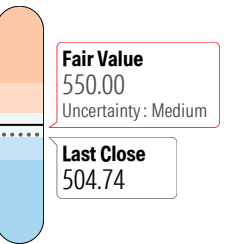
- There is still plenty of runway for growth in electronic payments.
- The scalable nature of the business could allow PayPal to improve its margins over time.
- PayPal's long-running experience in online payments is a unique asset that is becoming more valuable as e-commerce becomes a bigger piece of the pie.

Bears Say Brett Horn, CFA, Senior Equity Analyst, 10 Apr 2026

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Competitors

	PayPal Holdings Inc PYPL	Block Inc Class A XYZ	Visa Inc Class A V	Mastercard Inc Class A MA
	 Fair Value 80.00 Uncertainty: High Last Close 50.39	 Fair Value 84.00 Uncertainty: Very High Last Close 71.90	 Last Close 326.85 Fair Value 323.00 Uncertainty: Medium	 Fair Value 550.00 Uncertainty: Medium Last Close 504.74
Economic Moat	Narrow	Narrow	Wide	Wide
Currency	USD	USD	USD	USD
Fair Value	80.00 4 May 2026 16:35, UTC	84.00 4 May 2026 16:35, UTC	323.00 4 May 2026 16:28, UTC	550.00 4 May 2026 16:24, UTC
1-Star Price	124.00	147.00	436.05	742.50
5-Star Price	48.00	42.00	226.10	385.00
Assessment	Undervalued 4 May 2026	Fairly Valued 4 May 2026	Fairly Valued 4 May 2026	Fairly Valued 4 May 2026
Morningstar Rating	★★★★ 4 May 2026 05:00, UTC	★★★★ 4 May 2026 05:00, UTC	★★★★ 4 May 2026 05:00, UTC	★★★★ 4 May 2026 05:00, UTC
Analyst	Brett Horn, Senior Equity Analyst	Brett Horn, Senior Equity Analyst	Brett Horn, Senior Equity Analyst	Brett Horn, Senior Equity Analyst
Capital Allocation	Standard	Standard	Standard	Standard
Price/Fair Value	0.63	0.86	1.01	0.92
Price/Sales	1.47	1.85	16.53	13.15
Price/Book	2.29	1.95	17.76	65.38
Price/Earning	9.32	34.20	28.60	28.67
Dividend Yield	0.56%	0.00%	0.77%	0.65%
Market Cap	45.33 Bil	42.83 Bil	621.58 Bil	445.83 Bil
52-Week Range	38.46—79.50	44.27—82.50	293.89—375.51	480.50—601.77
Investment Style	Mid Value	Mid Growth	Large Blend	Large Blend

- PayPal is increasingly coming into competition with larger companies and recent results suggest its market share is slipping.
- Alipay and WeChat provide examples of how governments could favor local players, and this could shut PayPal out of a number of emerging market opportunities.
- Opportunities to monetize Venmo could be limited.

Economic Moat Brett Horn, CFA, Senior Equity Analyst, 10 Apr 2026

Payment processing of any type is highly scalable, as once a payment platform is established, there is little incremental cost per transaction. If viewed through the lens of the acquiring industry, PayPal has material scale with over \$1.8 trillion in annual volume, but falls short of the volumes handled by players such as Fiserv and Worldpay. However, the acquiring industry contains niches, and within the e-commerce space, PayPal is a clear leader, which we think places a narrow moat around PayPal's

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operations.

However, PayPal's branded business is not an acquirer; its unique model centers on a two-sided platform, with PayPal maintaining relationships with both merchants and consumers. We think this approach has material benefits and was instrumental in allowing PayPal to develop its current foothold in the industry. Having information on both sides of the transaction gives PayPal a meaningful edge in combating fraud, which was a very meaningful factor in the early days of the internet, and remains a key issue. With an ability to combat fraud, PayPal was in a position to become a valued partner on both sides of a payment. Further, the relative ease of using PayPal in an online transaction materially boosts conversion rates, with PayPal transactions converting at a rate of almost 90%, compared with an industry average of about 50%. This highlights the ease of checkout for consumers and the attraction for merchants. With these dynamics in place, PayPal was able to generate a network effect.

We typically think of a network effect as a very strong source of advantage, and one that often gives rise to a wide moat. In PayPal's case, though, we think the effect is much milder, as the company is a relatively small piece of the overall electronic payment infrastructure. We think PayPal will remain a preferred partner in the online world, given the relative convenience and security of its platform, but its market position is not so strong that the company can dictate terms to other players or gobble up increasing amounts of market share. In our view, the limitations of the network effect are most apparent in the company's decision years ago to switch to neutrality in terms of payment choices (that is, to allow consumers to choose their method of funding, as opposed to being defaulted to funding options that are more advantageous for PayPal).

Outside of its legacy PayPal business, we think its Braintree operations also benefit from a narrow moat due to scale-based cost advantages. While Braintree is smaller than competitors Stripe and Adyen, and volume growth is currently constrained by the shift toward more profitable business, it has grown quickly historically and looks to have carved out a position as a long-term leader. We think Venmo benefits from a network effect. However, moats in this area are somewhat speculative, given the lack of profitability and uncertain long-term economics.

Given the limited capital needs of the business, historical returns on invested capital have been dramatically higher than any reasonable estimate of the cost of capital.

Fair Value and Profit Drivers Brett Horn, CFA, Senior Equity Analyst, 10 Apr 2026

Our fair value estimate for PayPal is \$80 per share, which equates to 15.3 times our projected 2026 adjusted earnings per share estimate.

Growth started slowing in 2022 as the company faced some headwinds, and management has faced difficulties reigniting growth. Going forward, we expect top-line growth to come under more pressure in

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2026, as management attempts to reset the business, and macroeconomic issues could potentially create additional issues. However, we think PayPal can return to mid-single-digit revenue growth over the long haul as it rides the secular shift toward electronic payments generally and e-commerce and mobile payments more specifically. Our projections result in a 5% revenue compound annual growth rate over the next five years, with growth picking up modestly in the back half of our projection period.

Margins came under pressure due to the loss of the high-margin eBay relationship, but management more than offset this with sizable cost reductions over the past couple of years. However, the company's comments suggest margins will be pressured in 2026, as management increases investments for growth. We expect margins (adjusted to exclude amortization and one-time expenses) to decline materially in 2026, but thereafter to improve modestly as the company finds a stable base and returns to modest growth. The net effect is adjusted margins at the end of our projection period are 18%, about 2 percentage points below the 2025 level.

We use a cost of equity of 9.6% and a weighted-average cost of capital of 8.8% in our valuation.

Risk and Uncertainty Brett Horn, CFA, Senior Equity Analyst, 6 Apr 2025

The payment processing industry is evolving, and it is possible that new competition and future disruption could significantly reduce the profitability PayPal can generate or cut the company out altogether. As the company's revenue is directly tied to revenue at its merchant customers and online spending tends to be relatively discretionary, PayPal is sensitive to macroeconomic conditions.

PayPal's international operations present currency and execution risk. Some governments have shown a preference for local payment processors, which could freeze PayPal out of certain markets.

The ultimate economics of the Venmo business are difficult to predict.

We see the company's largest environmental, social, and governance risk as data security. Any company involved in processing payments has potential exposure to breaches in its systems.

Our Morningstar Uncertainty Rating is High. However, this uncertainty is balanced, and we believe PayPal is as likely to materially outperform our long-term expectations as it is to underperform.

Capital Allocation Brett Horn, CFA, Senior Equity Analyst, 10 Apr 2026

Our Morningstar Capital Allocation Rating for PayPal is Standard. In our opinion, the company's balance sheet is sound, its capital investment decisions are fair, and its capital return strategy is appropriate.

We think management's strategic choices and its acquisitions have generally been sound. Former CEO Dan Schulman joined eBay in 2014 and led PayPal following its spinoff in 2015. While we don't have any major issues with Schulman's leadership, he made some significant mistakes, and the company was entering a new phase. As such, we think a change in management was the right move.

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Alex Chriss took over as CEO in September 2023 and he seemed like a solid choice. We believed it would be difficult to veer from the company's focus on improving margins and returning capital to shareholders, and we think that largely proved correct. But the board thought that execution was lacking and announced that Enrique Lores (currently serving as chair) would take over as CEO on March 1. Lores has had a long career at HP and has served as CEO there since 2019. We think his experience managing a tech company and familiarity with PayPal could be positives in accelerating PayPal's growth as quickly as possible, although 2026 looks set to be a transition year.

We think PayPal's management has historically successfully navigated the ongoing evolution of the electronic payments space, and its choices have helped build and maintain the narrow moat that surrounds the business. While PayPal has arguably simply been better placed for growth given its online focus, we think management did a good job exploiting the opportunities in front of the company and working to minimize competitive threats. We would point to a couple of factors to justify our favorable view. We think the switch to consumer funding choice was a savvy move and suggests management understands both the basis of the company's moat and its limitations. This move positioned PayPal as more of a partner to other players in the ecosystem and headed off competition that might have otherwise arisen.

That said, the company has run into some issues in recent years. Active account growth stalled starting in 2022. Management noted that incentive programs during the pandemic to add accounts proved to have relatively poor economics, as accounts acquired through incentives tended to churn quickly. As a result, management has shifted investment to focus on engagement with existing accounts. While this switch makes strategic sense to us, we think the situation highlights the uncertainty that accompanies PayPal's growth. Further, the company's inability to reignite growth over the past few years points to deeper issues management has yet to solve.

Since its spinoff, PayPal has been fairly active in terms of mergers and acquisitions. We like the iZettle deal, from a strategic perspective. Purchased for \$2.2 billion, iZettle is a European company with a business model similar to Square. We think the addition of this company gives PayPal a solid platform to expand into card-present transactions, a move we view as timely given the increasing convergence of online and brick-and-mortar acquiring. Among PayPal's other acquisitions, we would question Xoom, which was purchased for \$900 million in 2015. We see little strategic connection between PayPal's core business and international money transfers. PayPal also acquired Honey for approximately \$4 billion. This deal made strategic sense, in our opinion, but the price tag was difficult to justify. On the whole, we don't think PayPal's acquisitions have been value-creative. Given the greater focus on capital return, we don't expect any large acquisitions anytime soon.

Analyst Notes Archive

PayPal: Shares Rise on Reports of Potential Takeover Brett Horn, CFA, Senior Equity Analyst, 23 Feb

PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

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2026

PayPal shares rose about 6% on reports that the company could be a potential takeover candidate. Why it matters: According to a Bloomberg report, PayPal has received unsolicited interest from potential buyers. Bloomberg reports that at least one entity is considering buying the entire company. Other buyers are interested in buying pieces of the company. While there are some synergies between the various parts of the business, the company certainly has some noncore operations, and we think a sale of even larger pieces of the business could be attractive at the right price. The bottom line: We will maintain our \$80 fair value estimate for the narrow-moat company. With the current stock price representing about a 14% free cash flow yield using management's expected adjusted 2026 free cash flow level, we think there could be room for a significant premium. While we have no details of a potential transaction and there can be no assurance that an actual offer will materialize, we see this news as supporting our view that shares are materially undervalued.

PayPal Earnings: Growth Slows, New CEO Appointed Brett Horn, CFA, Senior Equity Analyst, 3 Feb 2026

PayPal had a weak end to the year, and the lack of progress in accelerating growth has prompted a CEO change, with current HP CEO Enrique Lores set to take over. The market has responded negatively, taking the shares down almost 20% in Feb. 3 trading. Why it matters: PayPal saw growth slow in the fourth quarter, with net revenue up only 3% year over year on a constant-currency basis. Branded checkout was the main culprit, with payment volume up only 4% (1% online). This marks a significant deceleration from recent results. The company did show some progress in accelerating unbranded volume growth, but this wasn't enough to be a material offset. We see transaction margin dollars (essentially gross profit) as the key metric for the company. Growth here weakened as well, with only 3% year-over-year growth (4% excluding interest rate impacts). The bottom line: We expect to lower our \$94 fair value estimate by about 15% after digesting these results but still see the shares as materially undervalued. The near term will be more difficult than we had expected, but we continue to think that PayPal's scale puts a narrow moat around the business. The current stock price represents about a 15% free cash flow yield using management's expected 2026 free cash flow level. We think this creates significant upside if the company can maintain modest growth long-term. Big picture: Lores, who had been PayPal's chair, will take over as CEO on March 1. Board member David Dorman has taken over as chair. We appreciate the board's willingness to quickly step in and view this move positively. While we think Alex Chriss' tenure had its positives, we believe a change is warranted. Lores has had a long career at HP and has been CEO there since 2019. We think his experience in managing a tech company and his familiarity with PayPal could be positives in accelerating PayPal's growth as quickly as possible, although 2026 looks set to be a transition year.

PayPal Earnings: Broad-Based Strength Drives Growth Brett Horn, CFA, Senior Equity Analyst, 28 Oct 2025

PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

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Overall results largely held up better than expected in the third quarter, as PayPal showed strength across its business. Why it matters: Top-line growth in the third quarter was solid, with net revenue increasing 6% year over year on a constant currency basis and payment volumes up 7%. Growth was relatively broad-based. PayPal-branded volume growth held steady sequentially at 8% (although US volumes picked up a bit), while Venmo and PSP (Braintree) growth picked up to 14% and 6%, respectively. PSP is starting to lap efforts to cull unprofitable business. Additionally, management pointed to a 20% increase in buy-now-pay-later volumes as a driver across the business. We continue to see transaction margin dollar (essentially gross profit) growth as the key metric, given PayPal's shifted to a focus on profitable growth. This metric was up 6% year over year, which was better than expected. While interest rates remain a modest headwind, this is being partially offset by higher customer balances. The bottom line: We will maintain our \$94 per share fair value estimate for the narrow-moat company and see PayPal's shares as undervalued. We think the market underestimates the strength of the company's long-term competitive position and its growth prospects. The company announced a partnership with OpenAI to embed its wallet into ChatGPT. While we are encouraged to see PayPal open up new paths for growth, we are somewhat skeptical of the impact agentic payments will have on the overall consumer payments space over time, as we see limited use cases. Management also announced that PayPal will start to pay a modest dividend, with the company targeting a 10% payout ratio. We are pleased to see management reinforce its commitment to capital return, which we see as the right move both now and in the long term.

PayPal Earnings: Mixed Quarter; Market Focuses on the Negatives Brett Horn, CFA, Senior Equity Analyst, 29 Jul 2025

PayPal's second quarter was mixed, in our view, but the market appears to be focused on the lack of improvement in volume growth. Why it matters: In terms of top-line growth, PayPal largely maintained recent trends in the second quarter. Total payment volume was up 5% year over year on a constant-currency basis, with PayPal volumes up 5% (6% last quarter, adjusted for Leap Day). While consumer spending appears to be holding up overall, management noted that it saw some softness in areas affected by the new tariffs. Unbranded volume was up 2%, in line with the previous quarter, and Braintree was flat. However, management noted that they will start to lap the culling of Braintree's unprofitable volume in the third quarter and expect to return to growth. PayPal has shifted to a focus on profitable growth and is emphasizing transaction margin dollar (essentially, gross profit) growth, which we see as the correct move. In the quarter, this metric was up 7% year over year, in line with the previous quarter. Excluding the impact of interest rates, growth improved slightly sequentially to 8%. The bottom line: We will maintain our \$94 per share fair value estimate for the narrow-moat company and see the shares as undervalued. We believe the company maintains a strong competitive position, and management is on the right track. PayPal saw strong margin improvement in the quarter, with adjusted operating margins up 130 basis points year over year. While we don't expect PayPal to

PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

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maintain this rate of improvement, we think recent results highlight the scalable nature of the business. Based on the company's performance, management raised its full-year transaction margin dollar growth guidance by 1 percentage point and raised the range of its adjusted earnings per share guidance by about 4%.

PayPal Earnings: Recent Trends Persist; Progress in Shifting to Profitable Growth Brett Horn, CFA, Senior Equity Analyst, 29 Apr 2025

In our view, PayPal's first-quarter results contained no major surprises and largely showed a continuation of recent trends. We think the company is making progress in shifting toward profitable growth. Why it matters: While reported constant-currency net revenue growth of 2% was not impressive, this was largely expected. Underlying volume growth showed the company continuing to execute on its strategy. PayPal-branded year-over-year volume was 4%, or 6% excluding leap day last year. The adjusted figure was in line with growth last quarter. The bottom line: PayPal is tracking reasonably well, in our view, and we will maintain our \$94 fair value estimate for the narrow-moat company. We continue to see the shares as undervalued from a long-term perspective, although we see potential for meaningful near-term headwinds if the economy goes into a downturn. We believe transaction margin dollar (essentially gross profit) growth is the key metric to assess the company's pivot toward profitable volume. Year-over-year growth in transaction margin dollars was 7% excluding interest income, or 8% excluding the impact of leap year last year. This represented modest acceleration from the fourth quarter on an underlying basis.

Trump Tariffs Could Lead to Negative Impacts for Payment Names Brett Horn, CFA, Senior Equity Analyst, 6 Apr 2025

While we see the direct impact of the tariffs proposed by the Trump administration on payment processors as limited, the indirect effects could be materially negative. Why it matters: We think the recently announced Trump tariffs could potentially have a material negative impact on consumer spending, which could create significant headwinds for the payments space in the near term. To the extent that the tariffs lead to inflation, there could be some offsetting positive impacts for the space, as revenue is tied to the size of purchases in many cases. However, we think the negative effects on consumer spending will likely be the dominant factor, with the names that are tied primarily to online purchases (which tend to be discretionary) or small merchants being the most exposed. The bottom line: After adjusting our near-term assumptions down to reflect a potentially more adverse macroenvironment, we are lowering the fair value estimates for the payment names that we see as the most exposed by about 10%. We will maintain our narrow moat ratings for these companies. We still see these names as materially undervalued from a long-term perspective, but investors may need to ride out an uncertain and difficult near term to get there. PayPal's online focus leaves it relatively exposed to an economic downturn, and we are lowering our fair value estimate on the firm to \$94 per

PayPal Holdings Inc

PYPL★★★★

4 May 2026 21:35, UTC

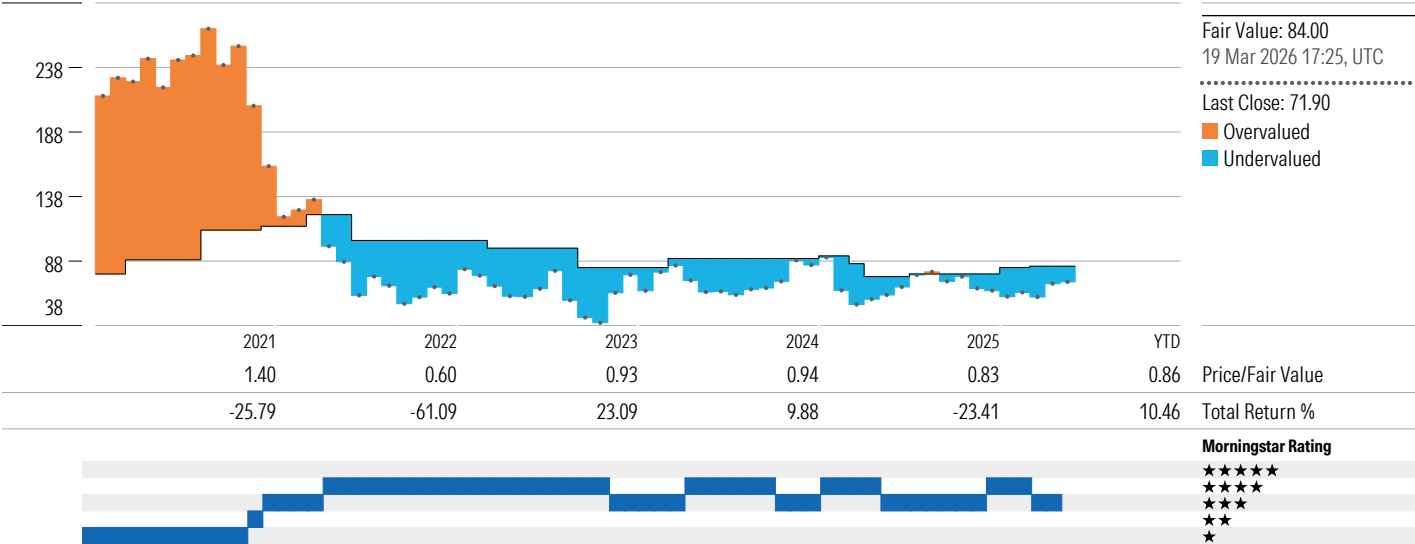
Last Price 50.39 USD 4 May 2026	Fair Value Estimate 80.00 USD 4 Feb 2026 01:26, UTC	Price/FVE 0.63	Market Cap 41.33 USD Bil 5 May 2026	Economic Moat™  Narrow	Equity Style Box  Mid Value	Uncertainty High	Capital Allocation Standard	ESG Risk Rating Assessment¹  5 May 2026 05:00, UTC
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share from \$104. Block and Global Payments' focus on small merchants leaves them relatively exposed as well. We are lowering our fair value estimate for Block to \$76 per share from \$86 and lowering our fair value estimate for Global Payments to \$146 per share from \$161. ■■

PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

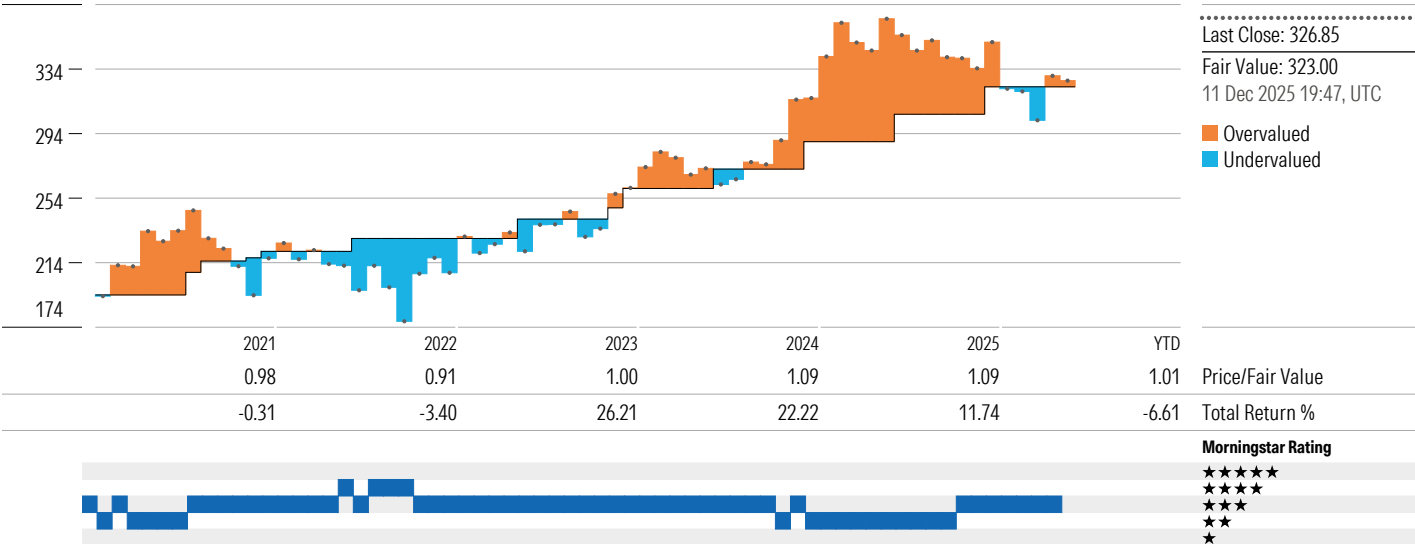
Competitors Price vs. Fair Value

Block Inc Class A XYZ



Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 19 Mar 2026 17:25, UTC.

Visa Inc Class A V

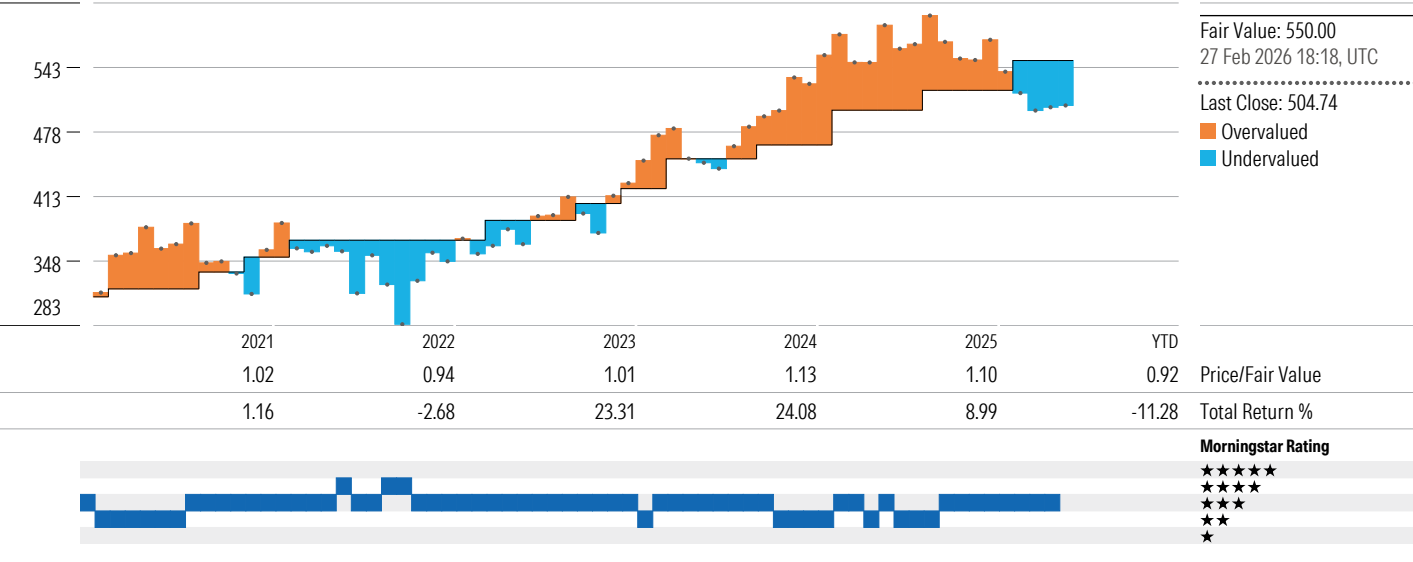


Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 11 Dec 2025 19:47, UTC.

PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

Competitors Price vs. Fair Value

Mastercard Inc Class A MA



Total Return % as of 04 May 2026. Last Close as of 04 May 2026. Fair Value as of 27 Feb 2026 18:18, UTC.

PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

Last Price 50.39 USD 4 May 2026	Fair Value Estimate 80.00 USD 4 Feb 2026 01:26, UTC	Price/FVE 0.63	Market Cap 41.33 USD Bil 5 May 2026	Economic Moat™ Narrow	Equity Style Box Mid Value	Uncertainty High	Capital Allocation Standard	ESG Risk Rating Assessment¹ 5 May 2026 05:00, UTC
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Morningstar Valuation Model Summary

Financials as of 10 Apr 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (USD Mil)	29,771	31,797	33,172	34,189	35,660	37,276	39,301	41,447
Operating Income (USD Mil)	4,944	5,763	6,396	5,641	5,884	6,150	6,681	7,253
EBITDA (USD Mil)	6,483	6,361	7,255	6,842	6,983	7,250	7,707	8,261
Adjusted EBITDA (USD Mil)	6,483	6,361	7,255	6,842	6,983	7,250	7,707	8,261
Net Income (USD Mil)	4,246	4,147	5,233	4,392	4,446	4,620	4,995	5,409
Adjusted Net Income (USD Mil)	4,679	4,270	5,846	4,744	4,781	4,837	5,184	5,608
Free Cash Flow To The Firm (USD Mil)	5,383	2,667	4,844	4,113	4,779	4,545	4,846	5,238
Weighted Average Diluted Shares Outstanding (Mil)	1,107	1,039	968	908	813	745	682	624
Earnings Per Share (Diluted) (USD)	3.84	3.99	5.41	4.84	5.47	6.20	7.33	8.68
Adjusted Earnings Per Share (Diluted) (USD)	4.23	4.11	6.04	5.22	5.88	6.50	7.60	8.99
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Margins & Returns as of 10 Apr 2026

	3 Year Avg	Actual			Forecast					5 Year Avg
		2023	2024	2025	2026	2027	2028	2029	2030	
Operating Margin %	18.0	16.6	18.1	19.3	16.5	16.5	16.5	17.0	17.5	17.2
EBITDA Margin %	—	21.8	20.0	21.9	20.0	19.6	19.5	19.6	19.9	—
Adjusted EBITDA Margin %	—	21.8	20.0	21.9	20.0	19.6	19.5	19.6	19.9	19.7
Net Margin %	14.4	14.3	13.0	15.8	12.9	12.5	12.4	12.7	13.1	12.7
Adjusted Net Margin %	15.6	15.7	13.4	17.6	13.9	13.4	13.0	13.2	13.5	13.4
Free Cash Flow To The Firm Margin %	13.7	18.1	8.4	14.6	12.0	13.4	12.2	12.3	12.6	12.5

Growth & Ratios as of 10 Apr 2026

	3 Year CAGR	Actual			Forecast					5 Year CAGR
		2023	2024	2025	2026	2027	2028	2029	2030	
Revenue Growth %	6.4	8.2	6.8	4.3	3.1	4.3	4.5	5.4	5.5	4.6
Operating Income Growth %	16.5	22.3	16.6	11.0	-11.8	4.3	4.5	8.6	8.6	2.6
EBITDA Growth %	16.9	38.4	-1.9	14.1	-5.7	2.1	3.8	6.3	7.2	2.7
Adjusted EBITDA Growth %	15.7	38.4	-1.9	14.1	-5.7	2.1	3.8	6.3	7.2	2.6
Earnings Per Share Growth %	37.3	83.6	4.1	35.4	-10.5	13.1	13.4	18.1	18.4	9.9
Adjusted Earnings Per Share Growth %	37.3	65.8	-2.8	46.9	-13.5	12.6	10.4	17.1	18.3	9.9

Valuation as of 10 Apr 2026

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Price/Earning	14.5	20.8	9.7	9.7	8.6	7.8	6.6	5.6
Price/Sales	2.2	2.7	1.6	1.3	1.3	1.2	1.2	1.1
Price/Book	3.2	4.3	2.8	2.5	2.4	2.4	2.3	2.2
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	10.1	13.0	7.5	6.6	6.4	6.2	5.8	5.4
EV/EBIT	13.2	14.4	8.5	8.0	7.6	7.3	6.7	6.2
Dividend Yield %	—	—	—	—	—	—	—	—
Dividend Payout %	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 10 Apr 2026

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
ROA %	5.2	5.1	6.5	5.5	5.5	5.7	6.1	6.4
ROE %	20.2	20.3	25.8	23.5	26.0	29.4	34.0	38.3
ROIC %	130.0	140.8	116.4	90.5	88.6	92.5	100.1	106.9

PayPal Holdings Inc PYPL ★★★★★

4 May 2026 21:35, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
50.39 USD 4 May 2026	80.00 USD 4 Feb 2026 01:26, UTC	0.63	41.33 USD Bil 5 May 2026	Narrow	Mid Value	High	Standard	5 May 2026 05:00, UTC

Financial Leverage (Reporting Currency)

Fiscal Year, ends 31 Dec	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Debt/Capital %	12.8	10.4	15.5	11.4	12.1	12.6	13.1	13.6
Assets/Equity	3.9	4.0	4.0	4.3	4.7	5.2	5.6	6.0
Net Debt/EBITDA	-0.7	-0.2	-0.1	-0.4	-0.1	0.1	0.2	0.3
Total Debt/EBITDA	1.5	1.6	1.4	1.5	1.5	1.5	1.5	1.5
EBITDA/ Net Interest Expense	—	—	—	17.1	14.8	14.6	14.8	15.1

Forecast Revisions as of 10 Apr 2026

Prior data as of 3 Feb 2026	2026		2027		2028	
	Current	Prior	Current	Prior	Current	Prior
Fair Value Estimate Change (Trading Currency)	80.00	80.14	—	—	—	—
Revenue (USD Mil)	34,189	34,349	35,660	35,886	37,276	37,820
Operating Income (USD Mil)	5,641	5,668	5,884	6,101	6,150	6,618
EBITDA (USD Mil)	6,842	6,888	6,983	7,222	7,250	7,735
Net Income (USD Mil)	4,744	4,839	4,781	4,958	4,837	5,325
Earnings Per Share (Diluted) (USD)	4.84	4.96	5.47	5.48	6.20	6.43
Adjusted Earnings Per Share (Diluted) (USD)	5.22	5.22	5.88	5.72	6.50	6.70
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00

Key Valuation Drivers as of 10 Apr 2026

Cost of Equity %	9.6
Pre-Tax Cost of Debt %	4.5
Weighted Average Cost of Capital %	8.9
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	3.0
Stage II Investment Rate %	17.8
Perpetuity Year	15

Additional estimates and scenarios available for download at <https://pitchbook.com/>.

Discounted Cash Flow Valuation as of 10 Apr 2026

	USD Mil
Present Value Stage I	12,218
Present Value Stage II	23,525
Present Value Stage III	25,655
Total Firm Value	61,398
Cash and Equivalents	14,752
Debt	9,987
Other Adjustments	0
Equity Value	66,163
Projected Diluted Shares	848
Fair Value per Share (USD)	80.00

PayPal Holdings Inc PYPL ★★★★★ 4 May 2026 21:35, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
50.39 USD 4 May 2026	80.00 USD 4 Feb 2026 01:26, UTC	0.63	41.33 USD Bil 5 May 2026	Narrow	Mid Value	High	Standard	5 May 2026 05:00, UTC

ESG Risk Rating Breakdown

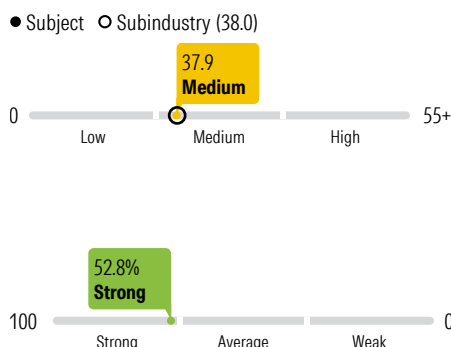
Exposure

Company Exposure¹	37.9
- Manageable Risk	34.8
Unmanageable Risk²	3.1

Management

Manageable Risk	34.8
- Managed Risk³	18.4
Management Gap⁴	16.4

Overall Unmanaged Risk 19.5



- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 52.8% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment⁵



ESG Risk Rating is of May 01, 2026. Highest Controversy Level is as of Apr 08, 2026. Sustainalytics Subindustry: Data Processing. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 05 May 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure		Management		ESG Risk Rating	
PayPal Holdings Inc	37.9 Medium	0 —●— 55+	52.8 Strong	100 —●— 0	19.5 Low	0 —●— 40+
Mastercard Inc	38.1 Medium	0 —●— 55+	62.0 Strong	100 —●— 0	16.3 Low	0 —●— 40+
Visa Inc	36.4 Medium	0 —●— 55+	66.6 Strong	100 —●— 0	13.9 Low	0 —●— 40+
Block Inc	33.9 Low	0 —●— 55+	47.5 Average	100 —●— 0	18.9 Low	0 —●— 40+
—	— —	0 —●— 55+	— —	100 —●— 0	— —	0 —●— 40+

Appendix

Historical Morningstar Rating

PayPal Holdings Inc PYPL 4 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★	★★★★	★★★★★	★★★★★	★★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★★	★★★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★★	★★★★★	★★★★	★★★★	★★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★★	★	★	★	★	★	★	★	★	★	★

Block Inc Class A XYZ 4 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★	★★★	★★★★	★★★★	★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★	★	★	★	★	★	★	★	★	★	★

Visa Inc Class A V 4 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★	★★★	★★★	★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★	★★	★★	★★	★★	★★	★★	★★	★★	★★	★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★	★★★★	★★★★	★★★★	★★★★	★★	★★	★★	★★	★★★★	★★	★★★★

Mastercard Inc Class A MA 4 May 2026 05:00, UTC

Dec 2026 —	Nov 2026 —	Oct 2026 —	Sep 2026 —	Aug 2026 —	Jul 2026 —	Jun 2026 —	May 2026 ★★★	Apr 2026 ★★★	Mar 2026 ★★★	Feb 2026 ★★★	Jan 2026 ★★★
Dec 2025 ★★★	Nov 2025 ★★★	Oct 2025 ★★★	Sep 2025 ★★	Aug 2025 ★★	Jul 2025 ★★	Jun 2025 ★★★	May 2025 ★★	Apr 2025 ★★★	Mar 2025 ★★★	Feb 2025 ★★	Jan 2025 ★★
Dec 2024 ★★	Nov 2024 ★★	Oct 2024 ★★★	Sep 2024 ★★★	Aug 2024 ★★★	Jul 2024 ★★★	Jun 2024 ★★★	May 2024 ★★★	Apr 2024 ★★★	Mar 2024 ★★★	Feb 2024 ★★	Jan 2024 ★★★
Dec 2023 ★★★	Nov 2023 ★★★	Oct 2023 ★★★	Sep 2023 ★★★	Aug 2023 ★★★	Jul 2023 ★★★	Jun 2023 ★★★	May 2023 ★★★	Apr 2023 ★★★	Mar 2023 ★★★	Feb 2023 ★★★	Jan 2023 ★★★
Dec 2022 ★★★	Nov 2022 ★★★	Oct 2022 ★★★★	Sep 2022 ★★★★	Aug 2022 ★★★	Jul 2022 ★★★	Jun 2022 ★★★★	May 2022 ★★★	Apr 2022 ★★★	Mar 2022 ★★★	Feb 2022 ★★★	Jan 2022 ★★★
Dec 2021 ★★★	Nov 2021 ★★★	Oct 2021 ★★★	Sep 2021 ★★★	Aug 2021 ★★★	Jul 2021 ★★	Jun 2021 ★★	May 2021 ★★	Apr 2021 ★★	Mar 2021 ★★	Feb 2021 ★★	Jan 2021 ★★★

Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, in-depth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss short-term market-price movements), but we believe these negatives are mitigated by deep analysis and our long-term approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our single-point star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a

long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in working capital accounts, and capital spending. Based on these projections, we calculate earnings before interest,

after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



Research Methodology for Valuing Companies

outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

Margin of Safety		
Qualitative Analysis	★★★★★ Rating	★ Rating
Uncertainty Ratings		
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

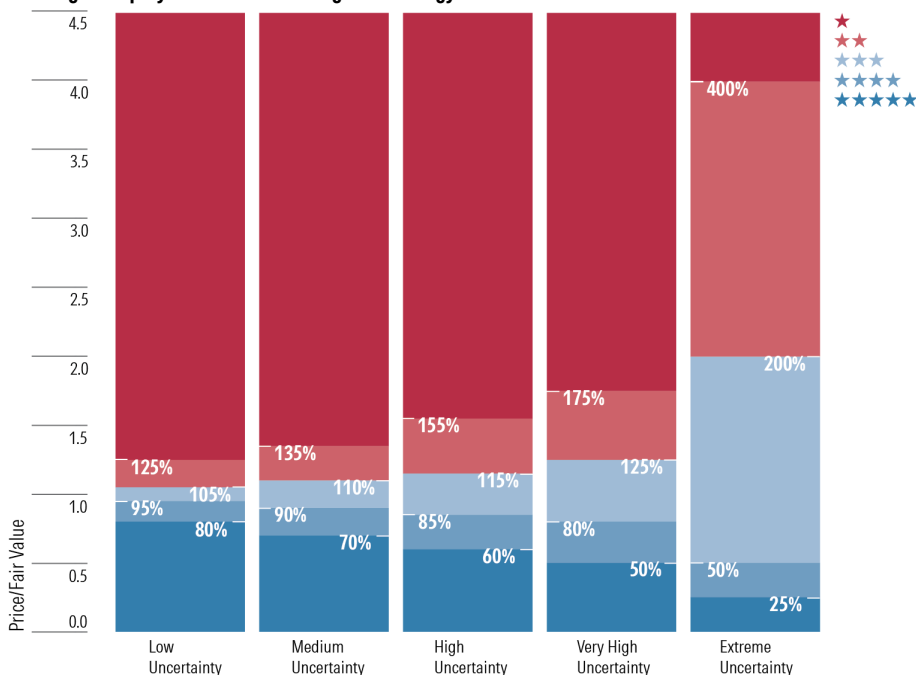
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to <https://shareholders.morningstar.com>

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk-adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multi-year time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

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Sustainalytics ESG Risk Rating Assessment: The ESG Risk Rating Assessment is provided by Sustainalytics; a Morningstar company.

Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score.

Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

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